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1.3.3 The Frobenius norm

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Week 01 due Oct 18, 2023 20:13 IST

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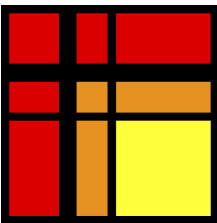
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Reading assignment

1/1 point (graded)



Read Unit 1.3.3 of the notes. [\[LINK\]](#)

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Homework 1.3.3.2

1/1 point (graded)
Prove that the Frobenius norm is a norm.

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Establishing that this function is positive definite and homogeneous is straight forward. To show that the triangle inequality holds it helps to realize that if $\mathbf{A} = \left(\begin{array}{c|c|c|c} \mathbf{a}_0 & \mathbf{a}_1 & \cdots & \mathbf{a}_{n-1} \end{array} \right)$ then

$$\begin{aligned} \|\mathbf{A}\|_F &= \text{< definition >} \\ &= \sqrt{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |\alpha_{i,j}|^2} \\ &= \text{< commutativity of addition >} \\ &= \sqrt{\sum_{j=0}^{n-1} \sum_{i=0}^{m-1} |\alpha_{i,j}|^2} \\ &= \text{< definition of vector 2-norm >} \\ &= \sqrt{\sum_{j=0}^{n-1} \|\mathbf{a}_j\|_2^2} \\ &= \text{< definition of vector 2-norm >} \\ &= \sqrt{\left\| \begin{pmatrix} \mathbf{a}_0 \\ \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_{n-1} \end{pmatrix} \right\|_2^2}. \end{aligned}$$

In other words, it equals the vector 2-norm of the vector that is created by stacking the columns of $\mathbf{A}$ on top of each other. One can then exploit the fact that the vector 2-norm obeys the triangle inequality.

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Homework 1.3.3.3

1/1 point (graded)
Partition $m \times n$ matrix $\mathbf{A}$ by columns:

$$\mathbf{A} = \left(\begin{array}{c|c|c} \mathbf{a}_0 & \cdots & \mathbf{a}_{n-1} \end{array} \right).$$

Show that

$$\|\mathbf{A}\|_F^2 = \sum_{j=0}^{n-1} \|\mathbf{a}_j\|_2^2.$$

☒ done



Solution:

$$\|\mathbf{A}\|$$

$$\begin{aligned} &\|A\|_F \\ &= \quad < \text{definition} > \\ &\sqrt{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |\alpha_{i,j}|^2} \\ &= \quad < \text{commutativity of addition} > \\ &\sqrt{\sum_{j=0}^{n-1} \sum_{i=0}^{m-1} |\alpha_{i,j}|^2} \\ &= \quad < \text{definition of vector 2-norm} > \\ &\sqrt{\sum_{j=0}^{n-1} \|a_j\|_2^2} \end{aligned}$$

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Homework 1.3.3.4

1.0/1.0 point (graded)

Let $A \in \mathbb{C}^{m \times k}$ and $B \in \mathbb{C}^{k \times n}$. Using what you once learned about matrix transposition and matrix-matrix multiplication, reason that $(AB)^H = B^H A^H$.



done



Solution:

$$\begin{aligned} &(AB)^H \\ &= \quad < X^H = \overline{X^T} > \\ &\overline{(AB)^T} \\ &= \quad < \text{you once discovered that } (AB)^T = B^T A^T > \\ &\overline{B^T A^T} \\ &= \quad < \text{you may check separately that } \overline{XY} = \overline{X} \overline{Y} > \\ &\overline{B^T} \overline{A^T} \\ &= \quad < \overline{X^T} = \overline{X}^T > \\ &B^H A^H \end{aligned}$$

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Homework 1.3.3.5

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times n}$.

$$\|A^H\|_F = \|A\|_F.$$

ALWAYS



 Answer: ALWAYS

For full solution see notes: [[LINK](#)]

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